

VT-PLANS Metagenomics Sequencing Center
245 Duck Pond Drive
Blacksburg, VA, 24061
Email: abdelrazek@vt.edu

SUBMISSION GUIDELINES FOR METAGENOME SEQUENCING

1. Submission Information

Submitters are required to fill out the provided submission sheet with all the requisite information pertaining to submitted samples. All columns highlighted in *red* are required. Nucleic acid submissions will require additional information, including sample quality. Failure to provide the necessary information can result in rejection of the submission, under the discretion of VT-PLANS staff.

2. Sample Labeling

All samples submitted to VT-PLANS are required to be labelled exactly according to the entered submission sheet. Each sample is required to be labelled using waterproof labels with the following information: **sample name**, **date** and **DNA/RNA** (for nucleic acid samples). Improperly labelled samples or samples not named in the submission sheet will not be accepted.

3. Nucleic Acid Sample Quantity and Quality

Submitted nucleic acid samples must pass baseline quantity and quality to be accepted for sequencing. Each sample must be accessed using both a Qubit fluorometer and Nanodrop Spectrophotometer prior to submission. Minimum quantity, per library type, are listed in Table 1. Ideal sample qualities listed in Table 2. Under the discretion of the submitter, samples below these standards may still be submitted for sequencing. In these case, VT-PLANS cannot guarantee sequencing results.

Table 1. Minimum sample quantity

Library Type	Total Qty (Qubit)	Ideal concentration
Genomic DNA (Ligation)	1000 – 2000 ng ¹	50+ ng/ul
Amplicons (PCR)	100 – 500 ng	~50 ng/ul ²
cDNA / RNA	500 – 1000 ng	N/A ³

1 – Genomic DNA should be of high molecular weight to improve sequence length

2 – Lower concentrations for smaller amplicon samples are acceptable. If you are unsure, please contact staff

3 – RNA/cDNAs in the sample MUST be intact

Table 2. Acceptable sample quality ranges

Quality ratio [†]	Type	Acceptable range
A260/A280	DNA	1.8-2.0
	RNA	2.0
A260/A280	Both	2.0-2.2

† - The ratio between the absorbance at 260nm, 280nm, and 230 nm as measure by a Nanodrop spectrophotometer.

4. Packaging and shipment

Tissue samples should be packaged and shipped according to the respective animal or plant clinic guidelines. Nucleic acid samples sent directly to VT-PLANS must be packaged with the following guidelines. All samples should be placed in a sealed, secondary container (e.g., zip-lock bag, test-tube rack) to ensure separation from packaging materials (e.g. ice) and reduce external contamination. For same day drop offs, samples are required to be **kept on ice**. Samples with shipment times longer than a day are required to **be frozen (at -20°C or -80°C)** to maintain sample quality during storage.

Drop-offs

Please contact VT-PLANS staff (See section 5) prior to drop-off of sample packages containing time-sensitive material or 10+ samples. All other

sample packages for VT-PLANS should be dropped-off between **9am -4pm** at:

CMRL 282
College of VetMed, Virginia Tech,
245 Duck Pond Drive,
Blacksburg, VA, 24061

Shipping

Sample packages may also be shipped directly to VT-PLANS. Packages should be labeled and shipped accordingly to Dr. Sahar Abdelrazek at:

Sahar Abdelrazek
Veterinary Medicine Hospital Stores
Phase III Dock C
245 Duck Pond Drive,
Blacksburg, VA 24061
United States

5. VT-PLANS staff contact

If there are any comments, questions, or issues with these guidelines, please contact VT-PLANS staff, **Dr. Sahar Abdelrazek at abdelrazek@vt.edu**.